

Michael Zandonella

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EDUCATION

Oregon State University

Bachelor of Science in Computer Science | 3.96 GPA

October 2023 - June 2026

Corvallis, OR

EXPERIENCE

Software Engineering Tutor

College of Engineering, Oregon State University

October 2025 - June 2026

Corvallis, OR

- Improved technical problem-solving for 5-15 students per term across data structures, algorithms, and full-stack software engineering projects by pair-programming through implementations and explaining concepts
- Expanded student understanding across Python, React, SQL, HTML, CSS, JavaScript, networking, security, debugging, and algorithmic problem solving by delivering targeted 1-on-1 technical instruction

E-Commerce Web Assistant

OSU Beaver Store

June 2024 - July 2025

Corvallis, OR

- Reduced back-stock item retrieval time by 55% by building a Python automation workflow that transformed catalog records and product images into printable inventory labels
- Improved e-commerce data quality across thousands of product listings and 500+ digital assets by auditing catalog records, resolving content inconsistencies, and identifying items for photography or online reactivation

PROJECTS

Rotations.lol | TypeScript, React, Node.js, Supabase, PostgreSQL, AWS SES, Docker

- Enabled 500+ users to monitor high-value discounts across 16,000+ in-game items with 99.9% platform uptime, helping users save an average of 46% off retail pricing by building and operating a full-stack sale tracking platform with live data ingestion and automated notifications
- Maintained continuously updated pricing and catalog data across ~80 new weekly sales by engineering a distributed ingestion pipeline that processes and stores live sale data in PostgreSQL
- Delivered reliable automated sale notifications across 6,100+ user-wishlisted items by developing a scheduled event-processing pipeline that matches live pricing updates, suppresses duplicate alerts, and automates email delivery through AWS SES and SNS

RFID-Based Asset Tracking System (Consultancy Project) | Node.js, React, Socket.IO, Express, SQLite, Docker

- Established real-time asset visibility and role-based tracking workflows for 400+ students and staff by translating stakeholder requirements into a full-stack system across frontend, backend, and RFID scanner components
- Achieved sub-second synchronization between RFID scanner hardware and a multi-user frontend by architecting an event-driven asset tracking system with resilient device recovery and a custom scanner SDK
- Accelerated development and stakeholder validation for a 5-person engineering team by building a mock scanner API and control interface that simulated RFID hardware interactions during 6 months of hardware unavailability

YouTube Transcript Search | Python, Elasticsearch, React, Flask

- Implemented timestamp-level search across arbitrary YouTube playlists with indexing throughput benchmarked at 4,400+ transcript segments per second by designing a full-stack retrieval system using Elasticsearch
- Improved transcript search precision and recall to a 0.955 NDCG@10 score across 15 evaluated search queries by implementing BM25-based ranking, exact phrase matching, fuzzy search, and multi-field query boosting
- Reduced long-form video search and navigation time from minutes to seconds across 250+ video playlists by building a scalable React and Flask transcript search platform with direct timestamped playback links

SKILLS

Languages: TypeScript, JavaScript, Python, SQL, HTML, CSS

Frontend & Backend: React, Node.js, Express, Flask, REST APIs, WebSockets

Databases & Search: PostgreSQL, MongoDB, Elasticsearch, Supabase

Infrastructure & Tools: Docker, Terraform, Ansible, Kubernetes, CI/CD, AWS, Nginx, Linux, Git